Exercise 1

SQL Foundations

SQL Fundamentals - SELECT AND FILTERING

01 SELECTING AND SORTING

a Retrieve all Coloumns from the employeestable

- Pulling data out of the table, removing duplicates, Sorting data.

```
SELECT *
FROM employee_table;
```

Employees table

ID	First_name	last-name	department	Salary	hire-date	City
01	Alice	Green	IT	7000	2020-01-10	Johanesburg
02	Brian	Lee	HR	45000	2019-03-22	Cape-Town
03	Cathy	Zulu	Finance	65000	2015-07-18	Durban
04	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
05	Eva	Naidoo	IT	72000	2017-09-30	Johanesburg

2. Find all unique department :

```
SELECT DISTINCT department
FROM employees_table;
```

Department
IT
HR
Finance
Marketing

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3 Retrieve first and last names ordered by salary descending?

```
SELECT first-name,
       lastname
FROM employees- table
ORDER BY salary DESC;
```

First-name	Last-name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4, Retrieve the top 3 highest- Paid employees?

```
SELECT Id,
       First-name,
       last-name,
       Salary
FROM employees- table
ORDER BY salary DESC
LIMIT 3;
```

ID	First-name	last-name	Salary
05	Eva	Naidoo	72000
01	Alice	Green	70000
03	Cathy	zulu	65000

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FILTERING WITH WHERE

5. Find employee in the IT department

```
SELECT ID,
       First-name,
       last-name,
       department
FROM employee-table
WHERE Department = "IT";
```

ID	First-name	last-name	Department
03	Alice	Green	IT
05	Eva	Naidoo	IT

6. Find employees in Finance with Sellary > ₹60000

```
SELECT Id,
       first-name,
       last-name,
       department
FROM employees-table
WHERE departme = "FINANCE" AND Sellary > 60000;
```

ID	First-name	last-name	Department	salary
03	Cathy	Zulu	Finance	65000

7. Find employees in the HR or Marketing department

```
SELECT Id,
       First-name,
       last-name,
       department
FROM employees-table
WHERE department = "HR" OR "Marketing";
```

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ID	First-name	last-name	department
02	Brian	Lee	HR
04	David	Mokoena	Marketing

8 Find employees not in IT?

```
SELECT Id,
       First-name
       Last name,
       department
FROM   employee-table
WHERE  NOT department = "IT";
```

ID	First-name	Last-name	department
02	Brian	Lee	HR
03	Cathy	Zulu	Finance
04	David	Mokoena	Marketing

9. Find employees in IT, HR, or Finance using IN?

```
SELECT Id,
       First-name,
       last-name,
       department
FROM employees-table
WHERE department IN ("IT, HR, Finance");
```

ID	First-name	last-name	Department
01	Alice	Green	IT
02	Brian	Lee	HR
03	Cathy	Zulu	Finance
05	Eva	Naidoo	IT

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10. Find employees in IT with Salary > 65000 and City Johannesburg?

```
SELECT Id,
       first_name,
       last_name,
       department,
       Salary,
       City
FROM employees_table
WHERE department = "IT"
AND   Salary > 65000
AND   City = "Johannesburg";
```

ID	First-name	Last-name	Department	Salary	City
01	Alice	Green	IT	70000	Johanne
05	Eva	Naidoo	IT	72000	Johanne

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